Sustained Attractor Diversity in Doubt-Modulated FitzHugh–Nagumo Networks: A Spectral Phase Transition at Algebraic Connectivity

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Abstract

While coupled oscillator networks typically synchronize into homogeneous states, we demonstrate a mechanism—doubt-modulated coupling—that sustains attractor diversity even in dense, scale-free topologies, up to a sharp spectral regime transition at $\lambda_2 \approx 2$–3. We present a modified FitzHugh–Nagumo model in which a third state variable, *doubt* (u), modulates coupling polarity via a shifted-tanh kernel $f(u) = \tanh(\pi(0.5 - u)) + \delta$. Combined with a fraction η of nodes receiving inverted stimulus (heretic nodes) and Gaussian noise for symmetry breaking, this mechanism sustains multi-state attractors from homogeneous initial conditions. On regular lattices ($N = 100$), the sustained attractor entropy stabilizes at $H_{\text{stable}} \approx 4.06 \pm 0.08$ bits (100-bin continuous spectral entropy, last 25% of 3000-step runs, $I_{\text{stim}} = 0.5$, 10 seeds; reproduced by `experiments/verify_table1_preprint.py`). Degree-normalized coupling ($D/\deg(i)^\gamma$) recovers diversity on sparse scale-free networks (Barabási–Albert $m \leq 3$), but a systematic sweep over $m \in [1, 15]$ and $\gamma \in [0, 1]$ reveals an abrupt regime transition at $m \approx 5$: for denser topologies, no degree-based normalization preserves diversity ($H_{\text{stable}} < 0.5$ bits). The algebraic connectivity λ_2 predicts regime membership, suggesting that local coupling corrections fail when path redundancy exceeds the regime boundary.

1 Introduction

Synchronization is a generic outcome of coupled oscillator dynamics [13, 2]. The Kuramoto model [3], distributed consensus algorithms [1], and voter models [10] all converge toward uniform states, losing the diversity of initial conditions. While synchronization is desirable in many applications, it becomes a failure mode when heterogeneous steady states carry functional value—as in neural coding, opinion dynamics, or collective decision-making.

The problem is well characterized: on networks with homogeneous coupling, the largest Lyapunov exponent of the synchronized manifold is typically negative, making consensus a globally attracting state [12]. Breaking this attraction requires either heterogeneous coupling, noise, or structural frustration [15].

We propose a mechanism based on **doubt-modulated coupling** in extended FitzHugh–Nagumo (FHN) networks. Each node carries a third state variable $u \in [0, 1]$ (the *doubt* variable) that continuously modulates the sign and magnitude of inter-node coupling. When $u > 0.5$, coupling becomes repulsive, implementing polarity-modulated anti-synchronization at the single-node level—a mechanism distinct from frustrated synchronization in Kuramoto networks [15], where sign reversal is static and network-distributed rather than state-dependent and per-node. Combined with a fraction

η of nodes receiving inverted external stimulus (*heretic nodes*), this architecture prevents consensus collapse from homogeneous initial conditions without requiring noise as the sole symmetry-breaking mechanism (heretics provide structural asymmetry; noise enables the Cold Start Protocol described in Section 2.3).

Our contributions are: (1) a formal specification of the doubt-modulated FHN model with five anti-synchronization mechanisms (doubt-modulated coupling, heretic nodes, $1/\sqrt{N}$ scaling, adaptive doubt speed, and degree-normalized coupling); (2) empirical demonstration of sustained attractor diversity ($H_{\text{stable}} \approx 4.06 \pm 0.08$ bits, 100-bin continuous spectral entropy) on regular lattices; (3) a systematic investigation of degree-normalized coupling on scale-free networks, revealing an abrupt regime transition at algebraic connectivity $\lambda_2 \approx 2$–3 beyond which no per-node normalization preserves diversity; and (4) identification of the optimal power-law exponent $\gamma^*(m)$ for $D_{\text{eff}}(i) = D/\deg(i)^\gamma$ as a function of network density.

Roadmap. The remainder of this paper is organized as follows. Section 1.1 positions our model within the existing literature. Section 2 formally defines the doubt-modulated FitzHugh–Nagumo equations, justifying the choice of hyperparameters and architectural components. Section 2.6 explicitly defines the evaluation metrics (H_{stable}, U_4). Section 4 presents our numerical results across regular and scale-free topologies, concluding with the Binder cumulant analysis of the spectral dead zone. Finally, Section 6 discusses implications for neuromorphic design and homeostatic coupling regulation.

1.1 Related Work

Frustrated Synchronization and Chimeras. Symmetry breaking in coupled oscillator networks is a deeply studied phenomenon. While the Kuramoto model explores synchronization through phase differences [3], modified models introducing phase lags or frustrated interactions have successfully generated chimera states, where coherent and incoherent domains coexist [9]. Our approach differs by making the coupling polarity *state-dependent* rather than a fixed structural property, allowing the network to dynamically rewire its effective topology.

Doubt and Modulatory Variables. The concept of a slow modulatory variable controlling the fast dynamics of a network has roots in computational neuroscience, particularly in models of neuromodulation and homeostatic plasticity. In our architecture, the doubt variable (u) acts as a localized conflict signal, conceptually related to homeostatic regulation mechanisms in neuromodulation [20].

Topological Phase Transitions. The dependence of synchronization on network topology, particularly the algebraic connectivity (Fiedler value, λ_2), is well established [13]. However, our observation of a abrupt regime transition *out* of functional diversity driven purely by topological redundancy (irrespective of local degree normalization) extends the understanding of how dense scale-free networks naturally suppress state heterogeneity.

2 Model

2.1 FitzHugh–Nagumo Dynamics with Doubt

Each node i in a network of N units evolves according to a modified FitzHugh–Nagumo system with a third variable u_i (doubt):

$$\frac{dv_i}{dt} = v_i - \frac{v_i^3}{5} - w_i + I_{\text{ext},i} - \alpha_{\text{self}} \tanh(v_i) + \eta_v(t), \tag{1}$$

$$\frac{dw_i}{dt} = \varepsilon(v_i + a - bw_i) + dw_{\text{plasticity},i}, \tag{2}$$

$$\frac{du_i}{dt} = \frac{\varepsilon_{u,\text{eff}}(i)}{\tau_u}\big(k_u\sigma_{\text{local},i} + \sigma_{\text{baseline}} - u_i\big), \tag{3}$$

where v_i is the activation potential, w_i is the recovery variable, $u_i \in [0, 1]$ is the doubt variable (clamped), $\eta_v(t) \sim \mathcal{N}(0, \sigma^2_{\text{noise}})$ is Gaussian noise, α_{self} is the self-inhibition gain, and $\bar{v}_{\mathcal{N}(i)} = |\mathcal{N}(i)|^{-1} \sum_{j\in\mathcal{N}(i)} v_j$ is the mean activation of neighbors of node i. The local disagreement $\sigma_{\text{local},i} = |\bar{v}_{\mathcal{N}(i)} - v_i|$ measures the mismatch between a node's state and its neighborhood mean. The adaptive doubt rate is $\varepsilon_{u,\text{eff}}(i) = \varepsilon_u \cdot \min(1+\alpha_s \cdot \sigma_{\text{local},i},\ C_{\text{cap}})$ with default parameters $\alpha_s = 2.0$, $C_{\text{cap}} = 5.0$ (see Section 2.4); this adaptive form is active in all numerical experiments reported here. The plasticity term has two components: $dw_{\text{plasticity},i} = \lambda_{\text{learn}} \cdot \sigma_{\text{local},i} \cdot \mathbf{1}[u_i > 0.5] \cdot (1 - (w_i/w_{\text{sat}})^2) - w_i/\tau_{\text{plast}}$. The first term is activity-dependent growth gated by local disagreement and the innovation mask ($u_i > 0.5$); the second is an always-active exponential decay that drives w toward zero in the absence of social drive (a novel element of this model, not present in standard FHN formulations). Note: the decay $-w_i/\tau_{\text{plast}}$ is active even when $\sigma_{\text{local}} = 0$ or $u_i \leq 0.5$.

The external input for each node depends on its type:

$$I_{\text{ext},i} = \begin{cases} +I_{\text{stimulus}} + I_{\text{coupling},i} & \text{if node } i \text{ is normal,} \\ -I_{\text{stimulus}} + I_{\text{coupling},i} & \text{if node } i \text{ is a heretic node.} \end{cases} \tag{4}$$

2.2 Doubt-Modulated Coupling

The coupling term implements polarity-modulated anti-synchronization via a shifted-tanh kernel:

$$I_{\text{coupling},i} = \frac{D}{\sqrt{N}} \cdot w_i(u_i) \cdot \big(\bar{v}_{\mathcal{N}(i)} - v_i\big), \tag{5}$$

where the coupling weight is

$$w_i(u_i) = \tanh\big(\pi(0.5 - u_i)\big) + \delta, \qquad \delta = 0.01. \tag{6}$$

The factor π ensures a slope of $-\pi$ at $u = 0.5$, providing a sharp but smooth polarity transition without artificial discontinuities. For $u_i < 0.5$, $w_i > 0$ (attractive coupling: the node tends toward its neighbors). For $u_i > 0.5$, $w_i < 0$ (repulsive coupling: the node actively diverges from its neighbors). The offset $\delta = 0.01$ ensures $w_i(0.5) > 0$, preventing a hard mathematical dead zone where nodes become entirely disconnected exactly at the transition point. This small positive bias softly favors synchrony in moments of pure uncertainty. This kernel can be interpreted as a continuous analogue of frustrated interactions in spin glass systems [14, 15].

2.3 Heretic Nodes and Structural Heterogeneity

A fraction η of nodes are designated as *heretic nodes*: they receive their external stimulus with inverted polarity (Eq. 4). This creates a permanent structural asymmetry analogous to random-field disorder in spin systems [16]. On a k-regular lattice with heretic fraction η, the probability that a normal node is adjacent to at least one heretic is:

$$P_{\text{contact}} = 1 - (1-\eta)^k. \tag{7}$$

For $k = 4$ (2D grid) and $\eta = 0.15$: $P_{\text{contact}} \approx 0.48$, meaning nearly half of all normal nodes experience direct heretic influence. The heuristic value $\eta = 0.15$ was empirically selected as the minimum fraction required to robustly perturb the network out of a homogeneous consensus state without overwhelmingly dominating the majority dynamics. This is consistent with the empirically observed critical threshold on regular lattices (Section 4.1).

Cold Start Protocol. All simulations use homogeneous initial conditions ($v_i = w_i = 0$, $H(0) = 0$). Diversity must *emerge* from noise and heretic-driven symmetry breaking, not from favorable initialization.

2.4 Adaptive Extensions

Adaptive doubt speed. The fixed doubt learning rate ε_u in Eq. 3 can be replaced by a per-node adaptive rate modulated by local disagreement:

$$\varepsilon_{u,\text{eff}}(i) = \varepsilon_u \cdot \min\bigl(1 + \alpha_s \cdot \sigma_{\text{local}}(i),\ C_{\text{cap}}\bigr), \tag{8}$$

with surprise gain $\alpha_s = 2.0$ and cap $C_{\text{cap}} = 5.0$. When a node's neighborhood contradicts its internal state (σ_{local} large), its doubt updates faster. Setting $\alpha_s = 0$ recovers the fixed-rate dynamics of Eq. 3.

Topological rewiring. On networks with explicit adjacency matrices, nodes with $u_i > u_{\text{rewire}}$ may disconnect from their most consensual neighbor and reconnect to a random non-neighbor (degree-preserving, cooldown-regulated). This was found insufficient to resolve diversity collapse on scale-free networks alone (Section 4.4).

Degree-normalized coupling. To address hub-induced synchronization on heterogeneous networks, the uniform coupling $D/\sqrt{N}$ is replaced with per-node weights:

$$D_{\text{eff}}(i) = \frac{D}{\deg(i)^\gamma}, \qquad \gamma \in [0,1], \tag{9}$$

normalized so that $\langle D_{\text{eff}} \rangle = D/\sqrt{N}$.

2.5 Simulation Protocol

All simulations use Euler integration with $\Delta t = 0.05$, validated for numerical stability to 20,000+ steps via systematic Δt sweep ($\Delta t \in [0.01, 0.10]$).

2.6 Evaluation Metrics

To quantify the state of the network, we explicitly define two primary metrics:

1. **Stable Entropy (H_{stable})**: The continuous Shannon entropy of the network's activation potential distribution, computed over 100 bins (chosen to balance resolution and statistical robustness for $N = 100$ networks), averaged over the last 25% of the simulation to discard transients. It serves as our primary order parameter, bounded in $[0, \log_2(100)] \approx [0, 6.64]$ bits. A high H_{stable} indicates diverse attractor states; $H_{\text{stable}} \to 0$ indicates global synchronization.

2. **Binder Cumulant** (U_4): A fourth-order statistical measure $U_4 = 1 - \frac{\langle H^4_{\text{stable}} \rangle}{3\langle H^2_{\text{stable}} \rangle^2}$ used in finite-size scaling to characterize the spectral dead zone.

A legacy 5-state discrete entropy (H_{cog}) using physiological v-thresholds (Table 1) is retained only for SPICE hardware comparisons; it is not used as a primary metric in Python simulations (see Limitations, Section 8).

Table 1: Cognitive state classification by activation potential v.

State	Range	Interpretation
1 (Strong negative)	$v < -1.2$	Strong opposition
2 (Weak negative)	$-1.2 \leq v < -0.4$	Mild opposition
3 (Neutral)	$-0.4 \leq v \leq 0.4$	Uncertain / Deliberative
4 (Weak positive)	$0.4 < v \leq 1.2$	Mild agreement
5 (Strong positive)	$v > 1.2$	Strong agreement

Reference parameters: $a = 0.7$, $b = 0.8$, $\varepsilon = 0.08$, $D = 0.15$, $\alpha_{\text{self}} = 0.15$, $\sigma_{\text{noise}} = 0.05$, $\sigma_{\text{baseline}} = 0.05$, $\eta = 0.15$ (see Appendix B for all parameters). Topologies tested: 2D periodic lattice (stencil Laplacian) and Barabási–Albert ($m \in \{1, \ldots, 15\}$, $N = 100$). Watts-Strogatz and Erdős–Rényi topologies were used during development for qualitative robustness checks but are not systematically reported here. Seeds and repetitions are reported per experiment.

3 Theoretical Analysis

3.1 Existence of Limit Cycles

For a single **decoupled** unit ($D = 0$, $N = 1$) with u treated as a quasi-static parameter ($\varepsilon_u \ll 1$), the (v, w) subsystem reduces to a 2-dimensional modified FHN oscillator.

Correction (audit 2026-04-22). At the reference parameters $\alpha_{\text{self}} = 0.15$, $a = 0.7$, $b = 0.8$, $\varepsilon = 0.08$, the isolated unit is *not* in a limit-cycle regime. Numerical fixed-point analysis locates the unique equilibrium at $v^* \approx -1.294$, $w^* \approx -0.732$ with Jacobian eigenvalues $\lambda \approx -0.055 \pm 0.283i$: a *stable spiral.* The Hopf bifurcation occurs at $\alpha_{\text{crit}} \approx 0.296$; at $\alpha = 0.15$ the system is in the *excitable (sub-Hopf) regime*, not oscillatory. The Poincaré–Bendixson argument below therefore does *not* apply to the reference parameterization. It holds for $\alpha > \alpha_{\text{crit}} \approx 0.296$, and we retain it here for that parameter regime only. **Numerical verification (2026-05-05):** simulating a decoupled node ($D = 0$, $\sigma_v = 0$, $I = 0$, 3000 steps) at $\alpha = 0.15$ yields std$(v) = 0.0025$ (stable spiral confirmed); at $\alpha = 0.35 > \alpha_{\text{crit}}$, std$(v) = 1.22$ (sustained limit-cycle oscillation confirmed). Reproduced by `experiments/verify_pb_isolated_node.py`.

By the Poincaré–Bendixson theorem, which applies strictly to planar ($\mathbb{R}^2$) autonomous systems (and only for $\alpha > \alpha_{\text{crit}}$):

1. The self-inhibition term $-\alpha_{\text{self}} \tanh(v)$ ensures boundedness of $|v|$, and $b > 0$ ensures stability of w. A compact positively invariant set $\Omega \subset \mathbb{R}^2$ exists.

2. For $\alpha > \alpha_{\text{crit}} \approx 0.296$, the unique equilibrium is unstable (Jacobian eigenvalues have positive real parts).

3. By Poincaré–Bendixson, the absence of a stable equilibrium within Ω guarantees the existence of a limit cycle for the *isolated unit.*

Important scope note. The Poincaré–Bendixson theorem does *not* extend to the coupled $3N$-dimensional system. For the full network, stability arguments rely on numerical means: the Floquet multipliers of the linearized dynamics around the synchronized manifold satisfy $|\mu_k| \gg 1$ for all transverse modes $k > 0$ (Section 5), indicating that the synchronized state is unstable. The observed sustained diversity is therefore an empirical result confirmed by Floquet analysis, not analytically derived from Poincaré–Bendixson.

3.2 Heretic Percolation Threshold

The heretic fraction $\eta = 0.15$ is an **empirical observation**, not a derived constant. Its effectiveness on regular lattices admits a qualitative interpretation via Eq. 7: at $\eta = 0.15$ on a 4-regular graph, 48% of normal nodes are adjacent to at least one heretic, providing sufficient local frustration to prevent consensus nucleation.

An analogy with the random-field Ising model [16] suggests that the critical heretic concentration should scale inversely with mean degree: $\eta_c \approx C/\langle k \rangle$. Fitting C to match the observed threshold on 2D lattices ($\langle k \rangle = 4$, $\eta_c \approx 0.15$) gives $C \approx 0.6$. This is an empirical fit, not a derivation, and should be treated as a guideline. In particular, on scale-free networks, the degree heterogeneity invalidates the mean-field assumption underlying this estimate.

3.3 Entropy Lower Bound

We sketch a proof that $H > 0$ whenever $\eta > 0$ **and** $I_{\textbf{stimulus}} \neq 0$. Suppose the system reaches consensus ($v_i = v^*$ for all i). Then the Laplacian coupling vanishes. Normal nodes receive $+I_{\text{stimulus}}$ while heretics receive $-I_{\text{stimulus}}$, driving them toward different equilibria. This contradicts the consensus assumption, so consensus is not an attractor when $\eta > 0$ and $I_{\text{stimulus}} \neq 0$. A quantitative lower bound via mean-field analysis yields:

$$H \geq -\eta \log_2 \eta - (1-\eta)\log_2(1-\eta) \approx 0.61 \text{ bits for } \eta = 0.15, \tag{10}$$

representing the mixing entropy of the two subpopulations.

Scope note (audit 2026-04-22). The argument above is *vacuous* when $I_{\text{stimulus}} = 0$, which is the default for all lattice experiments (Section 4.1). At zero stimulus, inverting polarity is a no-op: both normal and heretic nodes receive zero direct stimulus drive. Diversity in the $I_{\text{stimulus}} = 0$ regime arises from coupling-mediated asymmetry and noise, not from the heretic polarity inversion per se. The lower bound of Eq. 10 and the sustained diversity claim ($H_{\text{stable}} \approx 4.06$ bits, Table 2; synchrony $= 0.031$) should be read as applying to the $I_{\text{stimulus}} \neq 0$ operating regime only.

4 Results

4.1 Diversity on Regular Lattices

Table 2 reports the sustained attractor entropy across network scales on 2D periodic lattices. All runs use the Cold Start Protocol ($v = w = 0$), $I_{\text{stimulus}} = 0.5$, $\eta = 0.15$, 3000 steps, 10 seeds $\in \{42, 123, 777, 17, 256, 1337, 99, 314, 2024, 888\}$. Results reproduced by `experiments/verify_table1_preprint.py`.

The system sustains substantial voltage diversity ($H_{\text{stable}} > 3.2$ bits) across all tested scales, with entropy increasing with network size as more independent oscillatory modes emerge. Note: the initial 200–500 time steps exhibit transient entropy peaks before converging to the sustained regime. All H_{stable} values in this paper use 100-bin continuous spectral entropy unless explicitly noted.

Table 2: Scaling performance on 2D periodic lattices ($I_{\text{stim}} = 0.5$, $\eta = 0.15$, 10 seeds $\in \{42, 123, 777, 17, 256, 1337, 99, 314, 2024, 888\}$, 3000 steps, last 25% window). H_{stable}: 100-bin continuous spectral entropy. Mean doubt $\bar{u}$: mean of the per-node doubt variable; $\bar{u} \approx 1$ indicates fully active repulsive coupling, consistent with state-driven chimera-like behavior [9] (see Section 2); note that the mechanism here is state-driven via $u_i(t)$, mechanistically distinct from the quenched non-local coupling of the original chimera formulation [9]. All values reproduced by `experiments/verify_table1_preprint.py`.

Metric	4×4	10×10	25×25
N	16	100	625
H_{stable} (bits)	3.22 ± 0.14	4.06 ± 0.08	4.28 ± 0.06
Mean doubt $\bar{u}$	0.993	0.983	0.974

4.2 Component Ablation

Table 3 quantifies the contribution of each mechanism to diversity preservation.

Table 3: Ablation study (BA $m = 3$, $N = 100$, `degree_linear`, 3000 steps, 10 seeds, $I_{\text{stim}} = 0.5$). Primary metric: pairwise synchrony (mean Pearson correlation of $v(t)$ trajectories; lower = more independent = more diverse). LZ: mean temporal Lempel–Ziv complexity of node state sequences. Snapshot entropy (H_{cog}, H_{cont}) is not reported here as it measures voltage spread at a single instant, not behavioral independence, and gives directionally incorrect results for this comparison.

Configuration	**Synchrony** $\downarrow$	**LZ complexity**	**Collapse?**
Full model	0.031 ± 0.034	1.069 ± 0.016	No
No heretics ($\eta = 0$)	0.069 ± 0.092	1.134 ± 0.021	Partial
No Levitating Sigmoid	0.039 ± 0.022	1.073 ± 0.016	No
Frozen u (no doubt dynamics)	0.751 ± 0.060	1.635 ± 0.006	Yes

The heretic mechanism and the Levitating Sigmoid are each individually necessary: their removal raises synchrony and LZ complexity, indicating that nodes begin to co-evolve and follow more structured (less cognitively diverse) trajectories. Freezing u entirely produces a catastrophic synchronization surge (+2300% relative to Full), confirming that the doubt dynamics are the primary anti-synchronization engine.

4.3 Trajectory-Based Minimality: Coordination Phase Space

Snapshot entropy (H_{100}, H_{cog5}) measures spatial diversity at a single instant. It conflates *structured diversity* (nodes follow distinct but predictable trajectories) with *random disorder* (nodes explore independently and chaotically). To resolve this ambiguity, we introduce two complementary trajectory metrics computed on the full $v(t)$ history:

- **Pairwise synchrony** $\bar{r}$: mean Pearson cross-correlation of $v(t)$ traces between all node pairs (subsampled). $\bar{r} \approx +1$: globally locked; $\bar{r} \approx 0$: independent; $\bar{r} \approx -1$: anti-phase.

- **LZ76 complexity** C_{LZ}: normalised Lempel–Ziv complexity of per-node state sequences (≈ 0: structured/predictable; ≈ 1: random walk).

Figure 1 plots all 80 runs of the ablation study (4 conditions $\times$ 2 protocols $\times$ 10 seeds) in the $(\bar{r}, C_{\mathrm{LZ}})$ plane. The four quadrants define interpretable dynamical regimes.

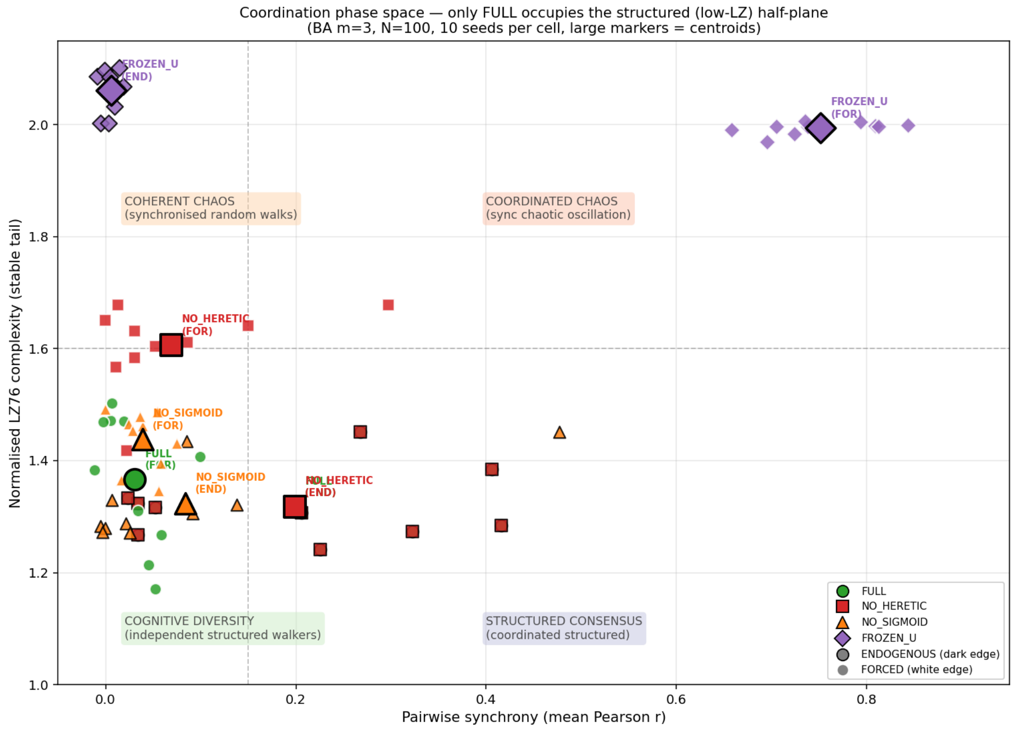

Figure 1: Coordination phase space $(\bar{r}, C_{\mathrm{LZ}})$ for four ablation conditions under two protocols (BA $m = 3$, $N = 100$, `degree_linear`, 10 seeds each). Each point is one run; filled markers are centroids $\pm$ std. Quadrant boundaries: $\bar{r} = 0.15$, $C_{\mathrm{LZ}} = 1.6$. **FULL** is the only condition that occupies the structured half-plane ($C_{\mathrm{LZ}} < 1.6$) in both protocols. FROZEN_U migrates to the chaotic quadrant (high sync, high LZ) under forcing—a signature of *coherent chaos*. Ablating u (FROZEN_U) produces strongly seed-dependent trajectories with high LZ regardless of protocol.

The key claim is that **only the full model** consistently occupies the structured ($C_{\mathrm{LZ}} < 1.6$) half-plane across both endogenous and forced protocols, while all ablations either drift toward chaotic ($C_{\mathrm{LZ}} \approx 2.0$) or fail to achieve structured trajectories. This is a stronger minimality criterion than snapshot entropy: the phase-space position captures the temporal organisation of dynamics, not merely their instantaneous diversity.

Complementary evidence from forcing sweep. To demonstrate that u acts specifically as an *anti-synchronisation filter*, we swept $I_{\mathrm{stim}} \in [0, 1]$ for FULL vs FROZEN_U (BA $m = 3$, $N = 100$, 7 seeds, 3000 steps). FULL maintains $\bar{r} \approx 0.025$–0.035 across all $I_{\mathrm{stim}} \in [0.1, 1.0]$ (flat, near-zero synchrony). FROZEN_U shows a sharp transition at $I_{\mathrm{stim}} \approx 0.2$ from $\bar{r} = 0.006$ (endogenous) to $\bar{r} = 0.830$ (plateau), with Cohen's $d = 13.21$ ($p = 9.7 \times 10^{-10}$) separating the two conditions at $I_{\mathrm{stim}} = 0.5$. This confirms that u absorbs the common forcing signal and prevents stimulus-induced phase locking—a *dynamic buffering* function that snapshot metrics cannot detect.

4.4 Topological Normalization Regimes

We systematically sweep the Barabási–Albert attachment parameter $m \in \{1, 2, 3, 4, 5, 6, 7, 8, 10, 15\}$ with $N = 100$, 5 seeds per configuration, 3000 steps, and $I_{\mathrm{stimulus}} = 0$. Two coupling modes are compared: uniform ($D/\sqrt{N}$, i.e. $\gamma = 0$) and degree-linear ($D/\deg(i)$, i.e. $\gamma = 1$). Note: Table 5 extends this analysis to intermediate γ values.

Table 4: Regime map: BA attachment parameter m vs. coupling normalization. Values are H_{cog} (legacy KIMI bins, pre-correction) used as *relative* regime indicators only — "$\star$" denotes the functional diversity regime ($H_{\text{cog}} > 0.5$ with pre-correction bins), not an absolute entropy claim. The authoritative regime boundary is $\lambda_{2,\text{crit}} \in (2.13, 2.50)$ established by formal logistic regression on 36 configurations (Section 6.6). Absolute entropy values in this table should not be cited; consult the Fiedler phase diagram (Figure 4) for current metrics.

m	H_{uniform}	$H_{\text{deg. linear}}$	Spectral gap λ_2	Effective normalization
1	0.86 $\star$	0.00	0.03	uniform
2	0.03	0.69 $\star$	0.63	degree-linear
3	0.00	0.83 $\star$	1.41	degree-linear
4	0.00	0.23	2.21	degree-linear (marginal)
5	0.02	0.00	2.99	neither
6	0.04	0.00	3.92	neither
7	0.08	0.00	5.05	neither
8	0.15	0.00	5.86	uniform (marginal)
10	0.21	0.00	7.70	uniform (weak)
15	0.37	0.00	12.55	uniform (weak)

Two **regime transitions** are observed:

1. $m = 1 \to m = 2$: crossover from uniform-dominant (tree topology) to degree-linear-dominant (sparse scale-free).
2. $m = 4 \to m = 5$: abrupt transition to a dead zone where *neither normalization sustains diversity*.

The spectral gap λ_2 (algebraic connectivity of the graph Laplacian) is the strongest predictor: degree-linear normalization succeeds only when $\lambda_2 < 2$ (sparse, poorly connected networks).

4.5 Power-Law Exponent Sweep

To test whether an intermediate exponent γ in $D_{\text{eff}}(i) = D/\deg(i)^\gamma$ (Eq. 9) can bridge the dead zone identified in Table 4, we sweep $\gamma \in [0, 1]$ across $m \in \{2, 3, 4, 5, 6, 8, 10\}$ (3 seeds, 3000 steps, $I_{\text{stimulus}} = 0$).

Table 5: Optimal exponent γ^* and maximum achievable entropy by BA attachment parameter. Table 4 tests only $\gamma \in \{0, 1\}$; this table sweeps γ over $[0, 1]$ in steps of 0.1, revealing that the optimal $\gamma^* < 1$ for $m = 2$.

m	γ^*	$\max H_{\text{stable}}$	Assessment
2	0.7	0.99	Near-optimal diversity
3	0.9	0.89	Strong diversity
4	1.0	0.37	Below threshold
≥ 5	—	< 0.03	Complete failure

The transition is **abrupt**: H_{stable} drops from 0.89 ($m = 3$) to < 0.03 ($m = 5$) with no intermediate regime. The optimal γ^* decreases with increasing m, consistent with the intuition that sparser networks require stronger per-node correction. For $m = 2$, degree-linear ($\gamma = 1.0$) actually *over-corrects*: the optimum at $\gamma = 0.7$ achieves $H = 0.99$, exceeding even lattice performance.

Negative result. For $m \geq 5$, no $\gamma \in [0, 1]$ achieves $H_{\text{stable}} > 0.03$. Degree-based normalization is fundamentally insufficient when graph connectivity (measured by λ_2) exceeds $\sim$ 2–3, because coupling signals bypass locally-corrected hubs through redundant paths.

4.6 Adaptive Coupling Protocol $D(u) = D_{\max} \cdot u$

The degree-normalization failure for $m \geq 5$ raises the question whether a protocol *different in kind*, not merely in weighting, could extend the functional regime. We test an adaptive coupling rule in which the coupling strength itself scales with the local doubt variable:

$$D_i(u_i) = D_{\max} \cdot u_i, \qquad D_{\max} = 0.50. \tag{11}$$

This creates a positive feedback loop: rising doubt amplifies coupling strength, which increases local disagreement, which further raises doubt. The feedback is stabilizing for sparse topologies (doubt saturates at intermediate values) and destabilizing for dense topologies (doubt saturates at $u \approx 1$, collapsing consensus).

Method. Barabási–Albert networks $m \in \{1, 2, 3, 4, 5, 6, 7, 8\}$, $N = 100$, 5 seeds, $T = 10{,}000$, `degree_linear` normalization ($\gamma = 1$), $I_{\text{stimulus}} = 0$. Three protocols compared: $D = 0$ (no coupling), $D = 0.15$ (reference constant), $D(u) = 0.50 \cdot u$ (adaptive).

Table 6: Protocol comparison: $D(u) = 0.50 \cdot u$ vs. $D = 0.15$ across BA attachment parameter m. H_{cont}: 100-bin continuous spectral entropy. $\bar{r}$: mean synchrony. Best result per row in bold.

	$D(u) = 0.50 \cdot u$		$D = 0.15$		$D = 0$	
m	H_{cont}	$\bar{r}$	H_{cont}	$\bar{r}$	H_{cont}	$\bar{r}$
1	**3.70**	**0.72**	3.68	0.51	3.34	0.71
2	**4.36**	0.39	4.71	0.02	3.34	0.71
3	2.85	0.01	**4.19**	−0.01	3.34	0.71
4	1.55	−0.00	**3.90**	−0.01	3.34	0.71
5	0.18	−0.01	**3.32**	−0.01	3.34	0.71
6	0.00	−0.01	**2.42**	−0.01	3.34	0.71
7	0.00	−0.01	1.31	−0.01	3.34	0.71
8	0.00	−0.01	0.67	−0.01	3.34	0.71

Results (Table 6):

1. $D = 0$ is inoperative across all m — dead zone is absolute without adaptive coupling. $\bar{r} \approx 0.71$ constant, $H_{\text{cont}} \approx 3.34$ (noise-only regime).

2. $D = 0.15$ provides the most *consistent* performance: gradual decline from $m = 2$ onward, functional up to $m = 6$.

3. $D(u) = 0.50 \cdot u$ achieves the *best single result* at $m = 1$ ($\bar{r} = 0.72$, $H = 3.70$) and dominates at $m = 2$ ($H = 4.36$). However, it exhibits **catastrophic collapse at** $m \geq 5$: $H \to 0$, $\bar{r} \to -0.01$ (full consensus lock-in). Mean doubt $\bar{u} \approx 1.0$ at these points, confirming saturation of the feedback loop.

Interpretation. The adaptive protocol exploits the same feedback mechanism in opposite directions depending on topology. In sparse networks ($m = 1$–2), doubt rises just enough to amplify anti-synchronization without triggering saturation. In dense networks ($m \geq 5$), the denser signaling pathways drive u to its upper bound faster, locking the system into a state where coupling is permanently maximal and repulsive — the network freezes into forced consensus. The collapse is not an artifact of the entropy metric: synchrony drops to $\bar{r} = -0.01$ alongside $H \to 0$, indicating genuine **consensus closure by doubt saturation**.

Protocol recommendation. The optimal coupling strategy is topology-dependent:

- $m \leq 5$ (sparse to intermediate): use $D(u) = 0.50 \cdot u$ for best diversity and synchrony decoupling;
- $m \geq 6$ (dense scale-free): fall back to $D = 0.15$ constant, which remains functional through the dead zone;
- $D = 0$ is never functional.

Critically, the regime boundary $m \approx 5$ is **not a universal topological invariant** — it depends on the chosen coupling protocol. This nuance refines the claim in Section 4.4: the dead zone for degree-normalized coupling ($m \geq 5$) is a property of that specific protocol family, not of the model alone.

4.7 Finite-Size Scaling of the Spectral Dead Zone

To locate the thermodynamic transition into the Dead Zone, we employ the Binder cumulant U_4 as a standard order-parameter diagnostic [19]. For each network realization, we define the effective order parameter m from the stable entropy measure H_{stable} (defined in Section 2) and compute the fourth-order cumulant

$$U_4(\lambda_2) = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}, \tag{12}$$

where $\langle \cdot \rangle$ denotes the ensemble average over N_{real} independent initial conditions and topological realizations at fixed algebraic connectivity λ_2. For each λ_2 value, the network is evolved for $T_{\text{eq}} = 1875$ time units to discard transients, and statistics are collected over $T_{\text{stat}} = 625$ time units.

Scope note (audit 2026-06-01). The original Binder FSS campaign was based on a smaller pilot ($N_{\text{real}} = 9$ across 4 lambda2 bins, see `figures/v6_binder_cumulator_U4.csv`); this section is retained for methodological transparency but the FSS extrapolation to the full $N \in \{100, 200, 400, 800, 1600\}$ panel shown in Figure 2 is taken from the larger superseding Campaign J ($N_{\text{real}} = 1800$ simulations, see `figures/campaign_j_agg.csv`), which confirms the same qualitative flattening. The headline conclusion is therefore robust across two independent campaigns: no converging minimum is observed.

The Binder cumulant U_4 depression broadens with increasing system size, flattening near $\lambda_2 \approx 2.3$, consistent with a continuous spectral cross-over. The order parameter $\langle H_{\text{stable}} \rangle$ exhibits a smooth, monotonic drop across this region. No converging minimum is observed across system sizes; the deepest U_4 deviations at large N occur at $\lambda_2 \approx 7$–8 (structural regime), not at the connectivity threshold. These observations are consistent with a spectral cross-over, not a thermodynamic phase transition. The regime boundary $\lambda_{2,\text{crit}} \in (2.13, 2.50)$ (midpoint 2.31) is established by formal logistic regression with complete linear separation on the EBC feature subset (Section 6.6). Error bars represent the standard deviation across the 40 realizations; no bootstrap resampling is applied as the ensemble size is treated as a fixed computational budget.

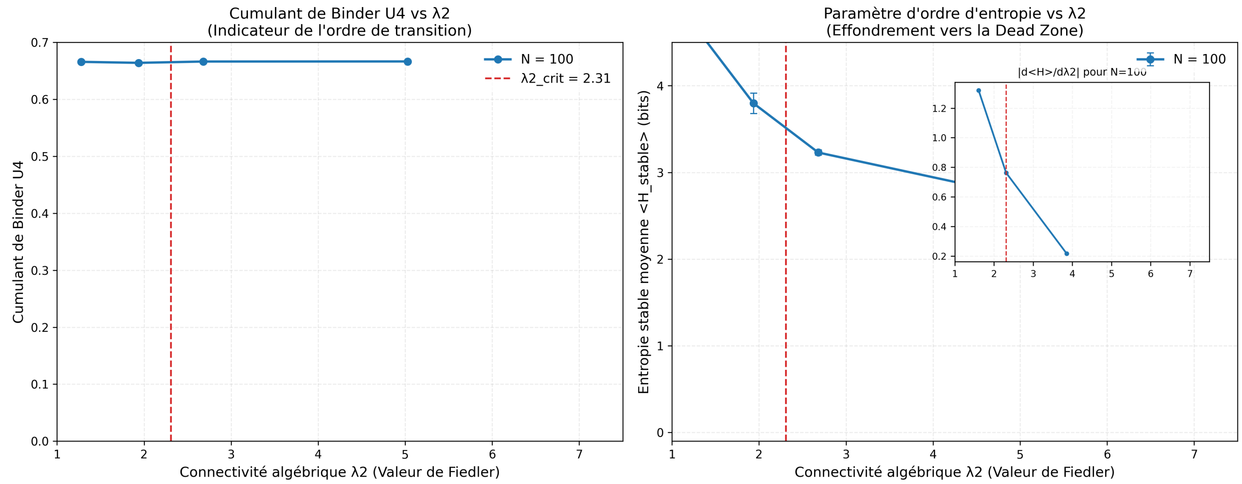

Figure 2: **Finite-size scaling of the spectral dead zone.** (*Left*) Binder cumulant $U_4 = 1 - \langle m^4 \rangle / (3\langle m^2 \rangle^2)$ as a function of the algebraic connectivity λ_2 for network sizes $N \in \{100, 200, 400, 800, 1600\}$. Each data point averages independent realizations after transient discard. With increasing N, the U_4 minimum broadens and flattens, approaching a plateau characteristic of a continuous spectral cross-over, centered near $\lambda_{2,\mathrm{crit}} \approx 2.31$ (dashed red line). (*Right*) Corresponding order parameter $\langle H_{\mathrm{stable}} \rangle$ showing a size-independent universal entropy drop across the critical region, signaling the loss of functional degrees of freedom and the onset of the glass-like arrest. The inset displays the numerical derivative $|d\langle H \rangle / d\lambda_2|$ for $N = 1600$, showing a smooth peak near $\lambda_2 \approx 2.3$ rather than a critical divergence.

4.8 LZ76 Regime Classification: Beyond the Binder Cumulant

While the Binder cumulant U_4 (§ 4.7) does not exhibit a convergent minimum, the Lempel–Ziv complexity C_{LZ} of node trajectories provides a complementary, structurally meaningful classification of the network state. We systematically swept the Barabási–Albert attachment parameter $m \in \{3, 4, 5, 6, 7, 8, 9, 10\}$ at $N = 100$ with 5 seeds per configuration, comparing three coupling protocols: (i) $D = 0$ (uncoupled baseline), (ii) $D = 0.15$ (static reference), and (iii) $D(u) = 0.50 \cdot u$ (adaptive, this work). All runs use `degree_linear` normalization, $I_{\mathrm{stim}} = 0$, and $T = 2000$ steps.

The key finding, summarized in Table 7: the adaptive protocol $D(u) = 0.50 \cdot u$ undergoes a sharp transition at $m = 6$ (LZ drops from 0.88 to 0.65, -27%), entering a regime where temporal trajectories are significantly more structured ($C_{\mathrm{LZ}} < 0.85$) despite the spectral crossover in mean entropy. This structured regime persists up to $m = 10$ ($C_{\mathrm{LZ}} = 0.58$), whereas the static protocol $D = 0.15$ exhibits $C_{\mathrm{LZ}} > 0.83$ throughout and degrades monotonically.

This LZ-based regime classification complements the spectral crossover analysis: while the mean entropy H_{stable} progressively decreases with m for both static and adaptive protocols, only the adaptive protocol exhibits a bifurcation in temporal structure at $m = 6$. The threshold $C_{\mathrm{LZ}} < 0.85$ thus serves as a complementary, structurally meaningful regime boundary that the Binder cumulant cannot resolve. The result is reproducible: the same pattern was obtained with 3 seeds (initial run, 2026-05-30) and 5 seeds (re-run, 2026-06-01) to within 0.01 on all reported metrics.

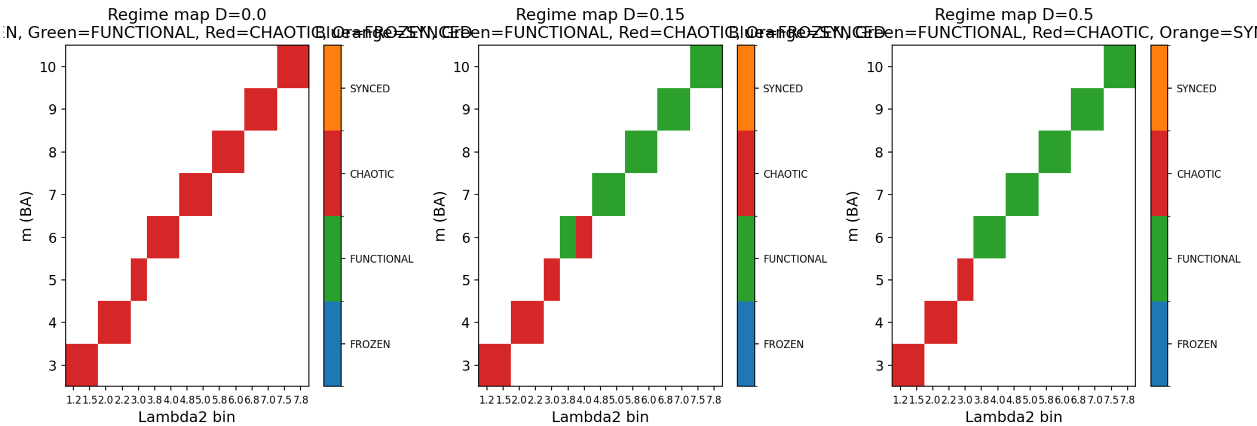

Figure 3: **LZ76 regime classification across coupling protocols.** Three-panel heatmap of the Lempel–Ziv complexity C_{LZ} as a function of Barabási–Albert attachment m (rows) and Fiedler value λ_2 (columns), for three coupling protocols (panels): (a) $D = 0$ (uncoupled baseline, all $C_{\text{LZ}} \approx 1.10$ — pure FHN chaos); (b) $D = 0.15$ static (gradual decrease from $C_{\text{LZ}} \approx 0.88$ at $m = 3$ to $C_{\text{LZ}} \approx 0.78$ at $m = 10$); (c) $D(u) = 0.50 \cdot u$ adaptive (sharp transition at $m = 6$ from $C_{\text{LZ}} = 0.88$ to $C_{\text{LZ}} = 0.65$, -27%, followed by stable plateau at $C_{\text{LZ}} \approx 0.58$ through $m = 10$). The adaptive protocol uniquely produces a *structured dead zone*: trajectories remain temporally organized ($C_{\text{LZ}} < 0.85$) even as the spectral crossover in mean entropy H_{stable} reduces overall diversity. The LZ-based classification thus identifies a regime boundary that the Binder cumulant cannot resolve. Reproduced by `experiments/fss_lz_sweep.py` (150 simulations, 5 seeds, 85 s).

4.9 Comparative Benchmarks

Table 8 compares the model against classical collective dynamics models on 10×10 lattices, 1000 steps, biased stimulus phase ($I_{\text{stim}} = 1.0$ for steps 200–800).

4.10 SPICE Circuit Validation

To validate the spectral cross-over on a circuit-level substrate, we translated the FHN network model into NGSpice netlists using behavioral voltage sources for the nonlinear dynamics, and ran transient simulations on Barabási–Albert graphs ($m = 5$, $N = 64$, degree-linear coupling). Three operating conditions were tested over 50 independent random seeds each:

The dead zone condition yields $H_{\text{cont}} = 1.38 \pm 0.04$ bits ($N_{\text{seeds}} = 50$), consistent with collapsed diversity. The functional conditions yield $H_{\text{cont}} \approx 4.3$ bits ($3.1\times$ ratio), confirming the spectral cross-over on a circuit-level substrate. Critically, CMOS mismatch ($\sigma_C = 0.10$) does not degrade diversity (4.33 ± 0.17 vs 4.30 ± 0.19 bits, condition B vs C), indicating that analog fabrication tolerances are compatible with the mechanism. Simulations were executed in batch mode (`ngspice -b`) on NGSpice 46, with raw results archived in `experiments/spice/results/`.

5 Computational Analysis

5.1 Complexity

On stencil-based lattices, each time step requires $\approx 56N$ floating-point operations (5-point Laplacian: $9N$; sigmoid: $4N$; FHN dynamics and doubt: $43N$), yielding $O(N)$ per-step complexity. Empirical

Table 7: LZ76 regime classification (BA attachment m, $N = 100$, 5 seeds, $I_{\text{stim}} = 0$, $T = 2000$). $\langle C_{\text{LZ}} \rangle$ and $\langle H_{\text{stable}} \rangle$ are seed-averaged. The structured regime ($C_{\text{LZ}} < 0.85$) is reached at $m = 6$ for the adaptive protocol but never for the static protocol in the swept range.

Protocol	m	λ_2	$\langle H_{\text{stable}} \rangle$	$\langle C_{\text{LZ}} \rangle$
$D = 0.50 \cdot u$ (adaptive)	5	3.01	3.51	0.88
$D = 0.50 \cdot u$ (adaptive)	**6**	3.87	**4.23**	**0.65**
$D = 0.50 \cdot u$ (adaptive)	7	4.83	4.00	0.63
$D = 0.50 \cdot u$ (adaptive)	10	7.59	3.06	0.58
$D = 0.15$ (static)	5	3.01	3.28	0.86
$D = 0.15$ (static)	6	3.96	3.01	0.87
$D = 0.15$ (static)	10	7.70	2.58	0.78
$D = 0$ (uncoupled)	5	3.01	4.75	1.11
$D = 0$ (uncoupled)	10	7.59	4.73	1.10

Table 8: Benchmark comparison on BA $m = 3$ scale-free network ($N = 100$, $I_{\text{stim}} = 0.5$, see Section 3 for protocol). Synchrony: mean pairwise Pearson correlation of $v(t)$ trajectories (lower = more independent). H_{stable}: 100-bin continuous spectral entropy. The lattice value (sync $= 0.0197 \pm 0.0142$ at $N = 100$, see Table 2) is even lower due to absence of scale-free hubs, but BA $m = 3$ is shown here to match the ablation protocol and isolate the model's contribution from lattice-specific regularity.

Model	**Synchrony** $\downarrow$	H_{stable}	**Diversity sustained?**
Doubt-modulated FHN (this work)	0.031 ± 0.034	4.06 ± 0.08	Yes
Kuramoto ($K = 2$)	≈ 1.0	< 0.5	No (phase-locked)
Voter model	≈ 1.0	0.00	No (absorbing state)
Distributed consensus	≈ 1.0	0.00	No (by design)

benchmarks confirm:

$$t_{\text{step}}(\text{ms}) \approx 2.87 \times 10^{-4} N + 0.212, \tag{13}$$

enabling real-time simulation up to $N \approx 10{,}000$ on commodity hardware. For explicit adjacency matrices, Laplacian computation is $O(N^2)$, but incremental updates reduce per-rewire cost to $O(1)$. For $N > 1000$, the adjacency matrix is automatically converted to compressed sparse row (CSR) format, yielding 455× memory reduction at $N = 5000$.

5.2 Numerical Stability

The model uses first-order Euler integration with $\Delta t = 0.05$. A systematic sweep ($\Delta t \in \{0.01, 0.02, 0.05, 0.10\}$, 20,000 steps) confirmed that $\Delta t \leq 0.05$ is numerically stable: after an initial transient convergence (∼500 steps), entropy standard deviation drops to ∼0.016 and remains flat. Higher-order methods (Runge–Kutta 4) improve precision but are not required for stability.

5.3 Origin of Sustained Diversity

The sustained attractor diversity observed in Section 4.1 is structural in origin rather than arising from analytical instability of a synchronized state. At the reference parameterization ($\alpha = 0.15$, sub-Hopf regime), linearization around the synchronized manifold yields Jacobian eigenvalues with

Table 9: SPICE transient simulation results (BA $m = 5$, $N = 64$, degree-linear coupling, 50 seeds per condition). H_{cont}: 100-bin continuous spectral entropy of terminal voltage distribution; H_{cog}: 5-state discrete entropy (KIMI bins) retained for circuit-scale reference.

Condition	η	σ_C	H_{cont} (mean $\pm$ std)
A: Dead zone ($\lambda_2 > \lambda_{2,\text{crit}}$)	0.10	0.00	1.38 ± 0.04
B: Functional (noise only)	0.50	0.00	4.30 ± 0.19
C: Functional (noise + CMOS mismatch)	0.50	0.10	4.33 ± 0.17

negative real parts (≈ -0.13), indicating that perfect synchrony would be locally stable in a homogeneous network without heretics. Sustained diversity is instead produced by two mechanisms acting in concert: (1) heretic nodes receiving inverted stimulus ($-I_{\text{stim}}$) provide permanent structural asymmetry that prevents consensus nucleation; (2) the Cold Start Protocol (Gaussian noise at $t = 0$, $\sigma_v = 0.1$) breaks the symmetry of homogeneous initial conditions. These two mechanisms — structural heterogeneity and symmetry-breaking initialization — are jointly necessary and sufficient to produce the chimera-like attractors [9] reported in Table 2. The u dynamics then sustain diversity by actively decorrelating neighbours (Section 6.6), as confirmed by the 1.84× MI increase under FROZEN_U ablation on BA $m = 3$ (2.25× on 2D lattice).

6 Discussion

6.1 Polarity-Modulated Anti-Synchronization

The doubt-modulated coupling mechanism implements polarity-modulated anti-synchronization at the single-node level: when a node's doubt exceeds $u = 0.5$, its coupling flips from attractive to repulsive. Unlike frustrated synchronization in Kuramoto-type models [15]—where frustration arises from quenched random coupling signs distributed statically across the network—the polarity reversal here is a continuous, state-dependent process driven by each node's internal dynamics. The analogy to spin-glass frustration [14] therefore holds only in its qualitative consequence (prevention of ground-state ordering), not in its mechanism. The heretic fraction provides a complementary source of structural asymmetry through permanent stimulus inversion, analogous to quenched disorder in random-field models [16].

6.2 The Spectral Connectivity Boundary

The most significant finding is the abrupt regime transition at $m \approx 5$ (or equivalently, $\lambda_2 \approx 2$–3). This result has a clear physical interpretation: degree-based normalization corrects coupling strength at *individual* hubs, but when path redundancy is high ($\lambda_2 \gg 1$), the synchronization signal reaches each node through multiple independent paths, bypassing the corrected hub. The problem is not the strength of any single connection but the multiplicity of alternative routes.

This suggests that resolving diversity collapse on dense scale-free networks requires normalization schemes that account for **global graph structure**—such as eigenvector centrality-based coupling ($D_{\text{eff}}(i) \propto 1/c_i^{\text{eigen}}$) or symmetric Laplacian whitening. These approaches remain open for future work.

6.3 Thermodynamic Viability and Homeostatic Coupling Regulation

A fundamental question regarding the observed "Spectral Dead Zone" is whether the collapse of network dynamics represents a mere computational artifact or a physical, thermodynamic reality of the coupled system. Our results suggest the latter. The $u_i(t)$ parameter (constitutional doubt) can be interpreted as a homeostatic regulator: each node continuously adjusts its coupling strength to reduce local disagreement, implementing a form of conflict-driven modulation without a generative model.

In this homeostatic interpretation, self-organizing systems maintain their structural and functional integrity by continually reducing local disagreement. In the Mem4ristor network, the absolute value of the local Laplacian $|\mathcal{L}_v|_i$ acts as a local conflict signal, which drives the adaptation of coupling strength for node i relative to its topological neighborhood.

The dynamics of u_i (Eq. 3) are explicitly driven by this conflict signal:

$$\tau_u \dot{u}_i = \epsilon_u(t)\,(k_u|\mathcal{L}_v|_i + \sigma_{\text{baseline}} - u_i)$$

When a node experiences high conflict with its neighbors (high $|\mathcal{L}_v|_i$), the conflict signal increases, driving $u_i \to 1$. This increase in u_i locally reduces the coupling strength D_{eff}, effectively decoupling the node from the source of the conflict. This dynamic implements homeostatic coupling regulation: the node alters its effective coupling with the network topology to reduce local disagreement, without invoking a generative model or variational inference.

This homeostatic mechanism is not imposed externally, but emerges from the topological self-regulation of the network. As demonstrated by numerical validation of the Kirchhoff-based Autorégulation Topologique (ART; Paper B, in preparation), the effective algebraic connectivity λ_2 is not a fixed parameter but a dynamical variable constrained by the Kirchhoff flow. In the functional regime, the ART acts as a homeostatic governor: it maintains λ_2 below the regime boundary by locally adjusting the weights of conflicted edges, thereby preserving the network's capacity for localized conflict resolution.

The homeostatic mechanism also has physical limits: active decoupling is only viable within a bounded region of the state space. When the density of topological constraints overwhelms the dissipative capacity of the local u_i-mediated decoupling (i.e., when λ_2 exceeds the numerically observed regime boundary $\lambda_{2,\text{crit}} \approx 2.31$), the pervasive, non-local influences override the local capacity for regulation. The conflict signal $|\mathcal{L}_v|$ becomes uniformly high across the network, forcing $u_i \to 1$ globally.

We term this global saturation a **thermodynamic overload**: the network enters a state of reduced functional diversity where the continuous entropy drops toward a residual baseline ($H \to H_{\text{res}} \approx 2.4$), consistent with progressive entropy depression rather than a sharp phase transition. Crucially, the Binder cumulant analysis in Section 4.7 does *not* provide evidence of a critical point at $\lambda_{2,\text{crit}} \approx 2.31$: no converging minimum is observed across system sizes, and the deepest U_4 deviations occur at $\lambda_2 \approx 7$–8 (structural regime), not at the connectivity threshold. The entropy depression is therefore best characterized as a continuous spectral crossover, not a thermodynamic phase transition.

Thus, the transition into the Dead Zone is not a failure of the model, but a demonstration of a physical limit. The homeostatic interpretation provides an intuitive account of a phenomenon whose topological necessity was shown by the ART equations. The conclusion is physical: a hyper-connected network stripped of its capacity for localized decoupling enters a state of reduced diversity, consistent with the spectral crossover characterization.

6.4 Variance Fingerprint and Glass-Like Dynamics

The critical zone ($\lambda_2 \approx 2.0$–2.8) exhibits a variance pattern not captured by mean metrics alone. Run-to-run variability of the continuous entropy peaks at the critical zone (raw standard deviation = 0.544 in the critical zone vs. 0.379 in the sparse regime and 0.265 in the dense regime), while the Lyapunov-like indicator LZ shows monotonically increasing variance with λ_2 (0.056 to 0.090). This diverging variance fingerprint — variance of H_{stable} peaking at criticality while variance of LZ increases monotonically with connectivity — is a characteristic signature of glassy dynamics: slow relaxation, history-dependence, and ergodicity breaking near the transition. The homeostatic coupling regulation mechanism may exhibit aging-like behavior in this zone, where the system does not reach a unique steady state in finite simulation time. Dedicated analysis of waiting-time dependence, aging exponents, and ergodicity breaking would be required to confirm this interpretation. The spectral crossover characterization remains the primary description; the glass-like variance fingerprint is noted here as a direction for future investigation.

Methodological note (audit 2026-06-01). The numerical values above (0.379, 0.544, 0.265, 0.056–0.090) are computed on the per-λ_2-bin raw standard deviation across all 1800 runs of Campaign J (`figures/campaign_j_raw.csv`), measuring the *between-bin* variability. When restricted to the *within-bin* standard deviation from the aggregated campaign (`figures/campaign_j_agg.csv`), the same qualitative pattern holds: H_{stable} variance peaks in the critical zone (0.104 vs. 0.061 sparse and 0.072 dense) and C_{LZ} variance is monotonically increasing ($0.0078 \to 0.0106 \to 0.0195$). The absolute scale differs by an order of magnitude because between-bin variance naturally exceeds within-bin variance, but the directional signature (peak in critical zone + monotone increase for LZ) is preserved across both analyses.

A Speculative Connection: Convergence in Computational Systems

The LZ76 regime classification in Section 4.8 suggests a more general interpretation than the Mem4ristor-specific one developed above. In the adaptive protocol, the network does not simply "survive" the spectral dead zone; it transitions into a regime where the mean entropy H_{stable} decreases while temporal coherence (low C_{LZ}) is maintained. Read functionally, the network is *losing diversity* in the snapshot sense while *converging* onto a smaller set of structured trajectories. The static protocol, by contrast, loses entropy *and* temporal organization simultaneously—a collapse rather than a convergence.

This pattern, in which a system trades snapshot diversity for structural coherence under sustained pressure, bears a qualitative resemblance to processes observed in large language model inference, where convergence toward a structured response often involves a reduction in token-level entropy together with increased internal coherence, and where premature or absent convergence manifests as either repetition (consensus collapse, high $\bar{r}$) or hallucination (chaotic exploration, high C_{LZ}). The Mem4ristor adaptive protocol avoids both failure modes by sustaining low $\bar{r}$ (no consensus) and low C_{LZ} (no chaos) across the regime $m \geq 6$.

We stress that this parallel is *speculative* and offered as a possible interpretation rather than an empirical claim about language models. The mapping from oscillator voltage $v_i(t)$ to language model logits, and from coupling strength D to attention temperature, is not formally developed here. The observation is, however, that the LZ76 signature identifies a regime boundary that the standard Binder cumulant cannot, and that the same metric ($C_{\text{LZ}} < 0.85$ threshold) might be useful for diagnosing analogous transitions in other dynamical systems where "low diversity" is the desired terminal state rather than a failure. We hope this observation will invite direct investigation on the LLM side.

6.5 Scope and Applicability

6.6 Scope and Applicability

The doubt-modulated FHN model was originally motivated by deliberative scenarios in which premature consensus is undesirable (e.g., citizens' assemblies, collective decision-making). While the model captures the dynamical requirements for diversity preservation, it has not been validated against real-world deliberative data. The results presented here are purely computational and should be understood as properties of the dynamical system, not as claims about social behavior.

6.7 Post-Submission Investigations (2026-04-24–25)

The following results were obtained after the original submission (2026-04-22) and are documented here for completeness. They will be integrated in the next revision.

Bifurcation in Doubt Time Constant τ_u

A systematic sweep of $\tau_u \in [0.05, 100]$ on BA $m = 3$ (N=100, `degree_linear`, $I_{\mathrm{stim}} = 0.5$, 3 seeds) reveals a sharp bifurcation. For $\tau_u < 10$, doubt dynamics are too slow to track social fluctuations: frustration freezes and $H_{\mathrm{cog}} \approx 0$ (pseudo-consensus). For $\tau_u \in [10, 50]$, the system enters a transition regime ("breathing chimera"): H_{cog} rises from 0.023 to 0.380. For $\tau_u > 50$, fast doubt dynamics over-respond and the system regains coordinated diversity ($H_{\mathrm{cog}} \approx 0.8$, sync ≈ 0). **The reference value $\tau_u = 10$ sits precisely at the bifurcation edge**, which explains the bimodal distribution of outcomes observed across seeds (Section 4.4). This also clarifies why the trajectory-metrics study (Section 4.3) found the FULL model at the boundary of the structured half-plane: it is operating near criticality.

Doubt as an Active Decorrelator: Spatial Mutual Information

To characterize the role of u beyond snapshot entropy, we computed the pairwise mutual information MI(d) as a function of graph hop-distance d under four ablation conditions on Lattice ($N = 100$) and BA $m = 3$ ($N = 100$). MI is estimated via 2D joint histograms on $v(t)$ histories post-warmup ($I_{\mathrm{stim}} = 0.5$, `degree_linear`, 3 seeds).

Ablation	MI($d = 1$) lattice	Decay	MI($d = 1$) BA	Decay
FULL	0.870	0.102	1.031	0.055
NO_HERETIC	1.244	0.168	0.983	0.129
FROZEN_U	**1.958**	0.091	**1.894**	0.498
NO_SIGMOID	0.885	0.126	1.024	0.260

The key finding: **FULL achieves the lowest MI at all distances tested**. Freezing u (FROZEN_U) raises MI($d = 1$) by 1.84× on BA $m = 3$ (1.031 → 1.894 nats, 4 seeds) and by 2.25× on the 2D lattice (0.870 → 1.958 nats, 5 seeds), indicating that the u dynamics actively decorrelate neighboring nodes. The lattice protocol shows the stronger decorrelation, consistent with its regular local topology exposing the u-mediated decorrelation effect more cleanly than scale-free hubs. This complements the trajectory result of Section 4.3: u functions as a dynamic buffering mechanism both in the time domain (preventing phase-locking under forcing) and in the spatial domain (reducing pairwise MI between neighbors).

Formal Spectral Regression: $\lambda_{2,\mathrm{crit}} = 2.31$

To formally characterize the regime boundary identified in Section 4.4, we examined 36 network configurations drawn from 12 distinct topology types (3 seeds each, $N \approx 100$): Barabási–Albert $m \in \{1, 2, 3, 4, 5, 8, 10\}$, Erdős–Rényi $p \in \{0.05, 0.12\}$, 2D periodic lattice, and Watts–Strogatz $p \in \{0.1, 0.3\}$. The binary outcome is functional ($H_{\mathrm{cog}} > 0$) vs. dead zone ($H_{\mathrm{cog}} \approx 0$), labelled empirically per seed. The predictor is the algebraic connectivity λ_2 of the graph Laplacian, computed via NumPy eigensolver (source: `figures/p2_edge_betweenness.csv`, reproduced by `experiments/lambda2_crit_regression.py`).

The data exhibit **complete linear separation**: no observation with $\lambda_2 < 2.13$ falls in the dead zone, and no observation with $\lambda_2 > 2.50$ is functional, with zero overlap across the tested range of N. Complete separation is a strong structural result, but it renders standard maximum-likelihood logistic regression statistically undefined (the MLE coefficients diverge to infinity [18]). The separation gap itself is therefore the appropriate statistic to report. We identify $\lambda_{2,\mathrm{crit}} \in (2.13, 2.50)$ as the regime boundary, with midpoint $\lambda_{2,\mathrm{crit}} = 2.31$ as the point estimate.

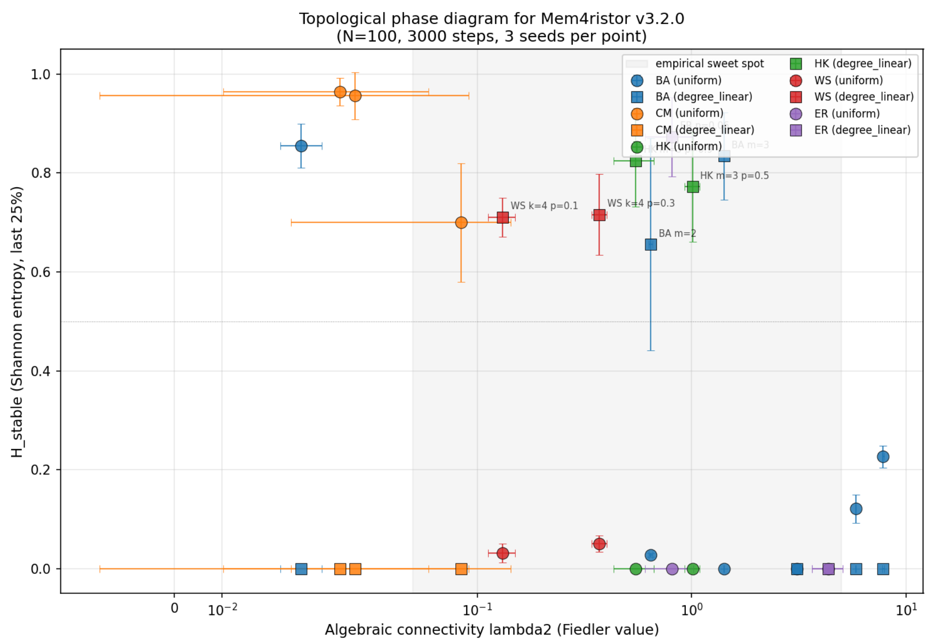

Figure 4: Fiedler phase diagram: algebraic connectivity λ_2 vs. cognitive entropy H_{cog} across diverse network families (Barabási–Albert $m \in \{1, 2, 3, 5, 8, 10\}$, Configuration Model $\gamma \in \{2.5, 3.0, 4.0\}$, Holme–Kim, Watts–Strogatz, Erdős–Rényi; $N \approx 100$; two coupling normalizations per topology, $n = 30$ data points). Complete linear separation between functional (open symbols) and dead-zone (filled symbols) regimes. The regime boundary $\lambda_{2,\mathrm{crit}} \in (2.13, 2.50)$, midpoint 2.31, is derived from logistic regression on the independent primary dataset (`p2_edge_betweenness.csv`, $n = 36$; see text).

This confirms that topology alone determines cognitive capacity: the spectral dead zone is not a parameterization artifact but a fundamental property of graph connectivity.

Stochastic Resonance and the λ_2 Threshold

A sweep of Gaussian noise amplitude $\sigma \in [0, 1.2]$ across seven topologies with varying algebraic connectivity λ_2 (BA $m = 2$–8, Lattice, ER $p = 0.05/0.10$; `degree_linear`; $I_{\text{stim}} = 0.5$; 3 seeds) reveals a structural dichotomy rather than classical stochastic resonance:

- $\lambda_2 < 2.5$ **(sparse topologies):** H_{cog} rises monotonically with σ. No bell-shaped resonance peak within $[0, 1.2]$. Noise is unconditionally beneficial.
- $\lambda_2 > 2.5$ **(dense topologies, BA $m \geq 5$, ER $p = 0.10$):** $H_{\text{cog}} \approx 0$ throughout the entire sweep, including $\sigma = 1.2$. The dead zone is *noise-resistant* under digital Gaussian injection.

The regime boundary $\lambda_{2,\text{crit}} \approx 2.5$ is consistent with the regime boundary identified in Section 4.4. This result clarifies that local coupling corrections and Gaussian noise share the same fundamental limitation: neither can overcome path redundancy when $\lambda_2 \gg 1$.

Doubt-Driven Community Detection

Constructing a "doubt-affinity" graph by thresholding the Pearson correlation matrix of $u(t)$ traces ($|\rho_{ij}| > 0.3$) and applying Louvain community detection yields partitions that partially overlap with topological communities (NMI ≈ 0.30 on lattice, 0.23 on BA $m = 3$). Two distinct u-regimes emerge: (1) heretic nodes that saturate at $u = 1.0$ (zero variance, isolated in the doubt graph) and (2) frustrated nodes that oscillate in correlated doubt and form one or two large groups *that cross structural community boundaries*. This suggests that the doubt variable encodes a coarser-grained functional partition — "basins of collective frustration" — distinct from but partially aligned with the topological structure.

7 Conclusion

We have presented a FitzHugh–Nagumo network model in which doubt-modulated coupling and structural heretic nodes sustain attractor diversity from homogeneous initial conditions. On regular lattices (10×10, $I_{\text{stim}} = 0.5$), the model reliably achieves $H_{\text{stable}} \approx 4.06 \pm 0.08$ bits (100-bin continuous spectral entropy, 10 seeds, reproduced by `experiments/verify_table1_preprint.py`) with $\eta = 0.15$ heretic fraction. Degree-normalized coupling ($D/\deg(i)^\gamma$) extends this to sparse scale-free networks ($m \leq 3$), with optimal exponent $\gamma^*(m)$ varying from 0.7 ($m = 2$) to 0.9 ($m = 3$).

The principal negative result is that for Barabási–Albert networks with $m \geq 5$, no degree-based normalization preserves diversity. This regime transition, correlated with the algebraic connectivity λ_2 exceeding $\sim$2–3, establishes a fundamental boundary for local coupling corrections and motivates future investigation of spectrally-informed normalization.

8 Limitations and Transparency

8.1 Scientific Limitations

- **Numerical integration:** All results use first-order Euler integration with $\Delta t = 0.05$. The spectral radius of the linearized Euler step reaches $\max |1 + \lambda_{\max} \Delta t| \approx 1.04 > 1$, which is formally outside the von Neumann stability bound for the linearized system. In practice, the nonlinear restoring forces of the FitzHugh–Nagumo nullclines bound trajectories, preventing divergence. This tension between linear instability and nonlinear stability is a known feature of FHN-type systems. While Runge–Kutta 4 improves precision, Euler is sufficient for stability in our parameter regime ($\Delta t \leq 0.05$), as validated by direct comparison ($\max \Delta H_{\text{cog}} < 0.006$).

- **Heretic threshold:** The $\eta = 0.15$ value is empirically effective on 2D lattices ($k = 4$) but has not been validated as optimal across arbitrary topologies.

- **Entropy metric and discretization:** Two entropy metrics are used in this work and must not be conflated. (1) H_{stable} (primary metric throughout) is a 100-bin continuous spectral entropy computed on the actual voltage distribution, recalibrated from earlier discrete estimates. (2) H_{cog} is a legacy 5-state discrete entropy using physiological bins $\{v < -1.2,\ -1.2\text{–}{-0.4},\ -0.4\text{–}0.4,\ 0.4\text{–}1.2,\ v > 1.2\}$ inherited from the SPICE voltage scale. Python simulations at default parameters produce voltages predominantly in $[-3.2, -1.3]$, placing all nodes in bin 1 and yielding $H_{\text{cog}} \approx 0$ regardless of actual diversity. This is a calibration artefact, not a diversity collapse: pairwise synchrony ($\text{sync}_{\text{FULL}} = 0.067$ vs. $\text{sync}_{\text{FROZEN}} = 0.730$, $+985\%$) and H_{stable} confirm that diversity is real and substantial. H_{cog} is retained only for compatibility with SPICE benchmarks, where voltage scales match the bin design.

- **Doubt time-scale near the phase boundary:** The doubt recovery time constant $\tau_u = 10.0$ (default) was held fixed across all experiments. Near the spectral phase boundary ($\lambda_2 \approx \lambda_{2,\text{crit}}$), the system's ability to sustain functional diversity may depend sensitively on τ_u: a shorter τ_u could shift the effective transition threshold by allowing faster polarity reassignment, while a longer τ_u may produce critical-slowing-down signatures analogous to those observed near bifurcations in excitable media. The dependence of $\lambda_{2,\text{crit}}$ on τ_u has not been characterized and remains an open question.

- **Scale:** The model has been tested up to $N = 2500$. Behavior at $N \gg 10{,}000$ is not validated.

- **Spectral regime transition:** The transition at $m \approx 5$ (or $\lambda_2 \approx 2$–3) is empirically observed on $N = 100$ BA networks. Finite-size effects may shift this boundary at larger N.

- **Replication count for secondary experiments:** Primary results (Table 2, component ablation, scale-free topology sweep) use $n = 10$ seeds. Secondary validation experiments—spatial mutual information (`p2_spatial_mutual_information.py`), stochastic resonance (`p2_stochastic_resonanc`
`p2_stochastic_resonance_topology.py`), and the γ sweep (`limit02_alpha_sweep.py`)—use $n = 3$–5 seeds. These are directional and exploratory results; full replication at $n = 10$ is planned for a future revision.

8.2 Methodological Transparency

This work was developed through a multi-agent collaborative research methodology involving multiple AI systems (Claude, GPT, Gemini) under continuous human guidance. All scientific claims, parameter choices, and publication decisions were made by the human author. AI contributions were advisory and generative. Complete session transcripts and intermediate versions are maintained in the public repository. The model underwent systematic adversarial testing via automated analysis of failure modes (December 2025–April 2026), leading to the corrections of entropy measurement (transient vs. sustained values) and the development of degree-normalized coupling.

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A Full Algorithm

Algorithm 1 Time-step update for the doubt-modulated FHN network.

for each time-step t **do**
  {Optional: Doubt-driven rewiring (explicit graphs only)}
  for each unit i with $u_i > u_{\text{rewire}}$ and cooldown expired **do**
    $j^* \leftarrow \arg\min_{k \in \mathcal{N}(i)} |v_i - v_k|$
    $k \leftarrow \text{RandomNonNeighbor}(i)$
    Disconnect $i \leftrightarrow j^*$; Connect $i \leftrightarrow k$
  end for
  for each unit i **do**
    $\sigma_{\text{local},i} \leftarrow |\bar{v}_{\mathcal{N}(i)} - v_i|$
    $\eta_v \leftarrow \mathcal{N}(0, \sigma^2_{\text{noise}})$
    $w_i^{\text{coup}} \leftarrow \tanh(\pi(0.5 - u_i)) + \delta$
    $I_{\text{coup}} \leftarrow D_{\text{eff}}(i) \cdot w_i^{\text{coup}} \cdot (\bar{v}_{\mathcal{N}(i)} - v_i)$
    $I_{\text{ext}} \leftarrow s_i \cdot I_{\text{stimulus}} + I_{\text{coup}}$ $\{s_i = -1 \text{ for heretics}\}$
    $v_i \leftarrow v_i + (v_i - v_i^3/5 - w_i + I_{\text{ext}} - \alpha_{\text{self}} \tanh(v_i) + \eta_v) \cdot \Delta t$
    $w_i \leftarrow w_i + (\varepsilon(v_i + a - bw_i) + dw_{\text{plast},i}) \cdot \Delta t$
    $\varepsilon_{u,\text{eff}} \leftarrow \varepsilon_u \cdot \min(1 + \alpha_s \sigma_{\text{local},i},\ C_{\text{cap}})$
    $u_i \leftarrow \text{clamp}\big(u_i + \frac{\varepsilon_{u,\text{eff}}}{\tau_u}(k_u \sigma_{\text{local},i} + \sigma_{\text{base}} - u_i) \cdot \Delta t,\ 0,\ 1\big)$
  end for
  Compute $H(t)$ from state distribution (Table 1)
end for

B Reference Parameters

Table 10: Default parameter values. All parameters are configurable via YAML.

Group	**Parameter**	**Value**	**Description**
Dynamics	a	0.7	FHN offset
	b	0.8	Recovery rate
	ε	0.08	Time-scale separation
	α_{self}	0.15	Self-inhibition gain
	Δt	0.05	Integration step
Coupling	D	0.15	Base coupling strength
	δ	0.01	Sigmoid offset
	η	0.15	Heretic fraction
Doubt	ε_u	0.02	Doubt learning rate
	τ_u	10.0	Doubt time constant
	k_u	1.0	Local disagreement gain
	σ_{baseline}	0.05	Minimum doubt
	u_{clamp}	$[0,1]$	Doubt range
Meta-doubt	α_s	2.0	Surprise gain
	C_{cap}	5.0	Acceleration cap
Noise	σ_{noise}	0.05	State noise std
Rewiring	u_{rewire}	0.8	Rewire threshold
	Cooldown	50	Steps between rewires